

GOOGLE'S SEWAGE COOLED DATA CENTERS

Believe it or not at its Georgia data centre, Google is using recycled water from toilets and bathtubs to cut down on cooling costs. Sounds yucky but quite environmental friendly considering data centers use thousands of gallons of water to cool their massive server farms
<http://dgit.in/daprw14>

WHY KONY WENT VIRAL

If you haven't seen the KONY 2012 video please crawl out from whatever rock you've been hiding under. Still, the most important lesson to be learnt here is to not believe everything you see on YouTube. And remember, we live in the age of online memes
<http://dgit.in/daprw12>

PIRATE BAY PLANS TO BUILD AERIAL SERVER DRONES



Digital pirates may soon have ships under their command. The popular torrent web site plans to move its proxy servers into the sky, creating a cluster of mobile computers tethered to GPS-enabled PCs
<http://dgit.in/daprw15>

ANDROID AND LINUX REUNITE!

Confused? When were they at loggerheads anyway? Turns out more than two years ago, Android code was booted from the staging area because no one was actually working to get it merged into the kernel. If Android is merged into the main tree, this would make it easier for Google to incorporate new Linux changes into its OS
<http://dgit.in/daprw16>

What to do with 88,128 parallel processors?

The short answer is: Not much. Or at least, nothing that mere mortals like you or I usually concern ourselves with. Because of the Linux OS, you wouldn't even be able to play Crysis 2 on a supercomputer — sheesh!

Supercomputers are mostly research tools. Historically (in the '70s) this was weather forecasting and aerodynamics (planes, space craft, cars), then nuclear weapons simulations and radiation shielding modeling (the height of the Cold War in the '80s), and in recent years the emphasis has been on cracking decryption and molecular dynamics modeling.

Let's break down a few of these applications so you can see why they're run on supercomputers.

With weather forecasting, it is the sheer amount of data that needs to be processed. There are hundreds of thousands of weather stations around the world, each one recording dozens of variables (wind speed/direction, temperature, humidity, pollen count). All of this data needs to be compared to last night's data, last week's data, and historical data, all of which must be fetched from huge memory and storage banks. Generally, most of these calculations can occur at the same time, and so massive parallelism means that weather forecasts can be produced in seconds or minutes.

With nuclear weapons simulation, scientists model every step that a nuclear bomb takes from storage in a silo to explosion. We're talking about a level of simulation that models whether the heat and vibration from a transport truck can affect a nuclear missile's control systems — but also the simulation of what actually happens when a nuke explodes. Is it better to explode it at higher altitude? What if we

throw in a bit more plutonium? And so on. Again, it's the sheer number of variables that makes massive parallelism such a boon here. One processor can work on the effects of humidity, while another can work on the fluid dynamics of fissile uranium.

Brute force decryption has fallen out of vogue in the last few years, but in the '90s you could use a supercomputer to break 56-bit DES encryption in a few minutes or hours. Here, massive parallelism simply lets you try different passwords at the same time; so if a single CPU can try a thousand passwords per second, a supercomputer with 10,000 CPUs can try 10 million passwords



per second. In general, though, brute force attacks take too long to be worth it — though that isn't to say that intelligence agencies around the world don't try (most supercomputers are state-owned, incidentally).

Finally we move on to by far the most exciting use of supercomputers: Molecular dynamics modeling. Basically, molecular dynamics is the simulation and analysis of the movements of atoms and molecules, and as such it covers the physical and mechanical aspects of almost everything in the known universe. To put this into perspective, Tianhe-1A, the second most powerful supercomputer in the world, ran a simulation involving 110 billion atoms through 500,000 time steps. In every one of these steps, Tianhe-1A has to analyze the

relationships between each and every atom. These calculations took three hours to complete and accounted for 0.116 nanoseconds of simulated time — and this is on a computer capable of processing two quadrillion calculations per second.

The most prominent example of molecular dynamics modeling Folding@Home, a distributed computing project that tries to work out the processes behind protein folding (incorrect protein folding is believed to cause a range of diseases, including Alzheimer's and cancers). Molecular dynamics is also used in pharmaceutical research — and indeed a lot of the world's largest, privately-

owned supercomputers are owned by Big Pharma. There has been a huge surge in molecular dynamics modeling since the completion of the first human genome project in 2000, and the many iterations since. In this area, supercomputers analyze how genes interact with one another, the effects of changing or deleting a gene, and the effect of gene therapy.

Molecular dynamics is also used to model various aspects of astronomy, including the first few seconds after the Big Bang and what happens when a star goes supernova. Molecular dynamics brings new levels of understanding to just about every chemistry or physics experiment, so it's not hard to see why everyone is rushing to build peta- and exascale supercomputers.